Vadim Vadanici

+33 6 47 54 95 05 | vadim.vadanici24@imperial.ac.uk | linkedin.com/in/vadim-vadanici | vvgit22.github.io

Education

Imperial College London — London, UK

MEng Biomedical Engineering (Computational Bioengineering Stream) — *Expected 2028*

- Predicted First Class Honours; top of a 120+ cohort in Year 1 Mechanics & Electronics (79% overall, 83% Mechanics).
- Undergraduate Researcher, Dario Farina Lab (2025–2026): machine learning on EMG hand-gesture recordings.
- Year 3 modules: Modelling in Biology; Digital Biosignal Processing; Probability, Statistics & Data Analysis.

École Active Bilingue Jeannine Manuel — Paris, France

French Baccalaureate, International Option (English) — *2024*

- Highest distinction (17.61/20) — perfect scores in Mathematics, AP Calculus BC, History, English and AP Language.
- Honourable Mention, French Mathematics Olympiad; Concours Général candidate (Philosophy, Physics-Chemistry).

Research & Healthcare Experience

PITOR — Paris Institute for Transplantation & Organ Regeneration — Paris, France

Research Intern — *August 2026–Present*

- Built a local LLM pipeline automating the thematic analysis of 6,000+ free-text responses from kidney-transplant patients, combining structured coding, embedding-based clustering and source-verified quotations.
- Benchmarked frontier LLMs against expert pathologists on Banff rejection diagnosis from lesion scores in 240 biopsies.
- Preparing a first-author manuscript, continuing part-time alongside my degree.

Okeiro (AI Medtech Start-up, Chronic Kidney Disease) — Paris, France

Summer Consulting Intern — *June–July 2025*

- Developed a US market-entry strategy through competitor benchmarking and nephrology market research.
- Designed a remote patient monitoring service model for chronic kidney disease.
- Pitched the business model to venture capital investors, reporting directly to the CEO.

French Institute of Health and Medical Research (INSERM) — Paris, France

Research Intern — *August 2023*

- Investigated BK virus pathophysiology and treatment approaches with a medical research team.

Engineering Projects

Neural Manifold Learning for Brain-Imaging Biomarkers — Imperial Group Project — London, UK

Project Manager (7-member team), supervised by Prof Simon Schultz — *October 2026–Present*

- Year-long project, the lab's first on human fMRI, applying manifold learning to the UK PREVENT Dementia cohort.
- Refining MIMI, an information-theoretic method that infers low-dimensional structure in brain activity.
- Analysing how default mode network manifolds change with age across participants to find Alzheimer's biomarkers.

Smart Elastic Resistance Bands for Physiotherapy — Imperial Group Project — London, UK

Project Manager (8-member team) — *October 2025–June 2026*

- Designed sensor wristbands that calibrate to each patient's range of motion, count reps on a resistance band and turn prescribed physiotherapy exercises into a video game to improve adherence at home.
- Sole developer of ReQuest: Bluetooth-driven game, patient app and clinician dashboard on a FastAPI backend.
- Led the team to the Year 2 Biomedical Engineering Prize for Best Group Project.

London AI x Science Hackathon (Iterate) — Polaron & Cognition Track — London, UK

One of 150 participants selected from 1,000+ applicants — *October 2026*

- Segmented battery-electrode microstructure via multi-Otsu to predict manufacturing changes before batteries are built.
- Linked literature-based markers that differed significantly between batches to a Doyle–Fuller–Newman battery model.

Certifications & Skills

Optiver & Imperial College London — Trading Academy: implemented trading strategies in Python *2026*

University of Cambridge — Machine Learning & AI for Medicine Summer School (van der Schaar Lab) *2025*

Programming: Python (NumPy, pandas, SciPy, scikit-learn, PyTorch), MATLAB, SQL, Java, Git

Software: SolidWorks, Excel, PowerPoint, LaTeX | **Languages:** Fluent in English, French and Romanian; German (B1)

Leadership & Extracurricular Activities

- Selected as one of 500 attendees from 5,000+ applicants for Y Combinator Startup School Paris (2026).
- Organiser, Nature Conference *Redefining Healthcare in the Age of AI* (Sorbonne, Paris, 2026).
- Imperial Neuroscience Society — selected for a mentorship programme led by neurotech founders.
- Former school basketball captain (10+ years); Imperial Basketball Team; EDUCAP/CAPSAA inclusive-sport organiser.
- Peer Leader and Class Delegate for international students; tutored Ukrainian students displaced by the war.
- Three exhibitions with Duchamp Prize winner Tarik Kiswanson; subject of his work *Vadim* (kadist.org/work/vadim).
- **Interests:** Neuroscience, artificial intelligence, music (DJing, piano), skiing, football, chess, philosophy.